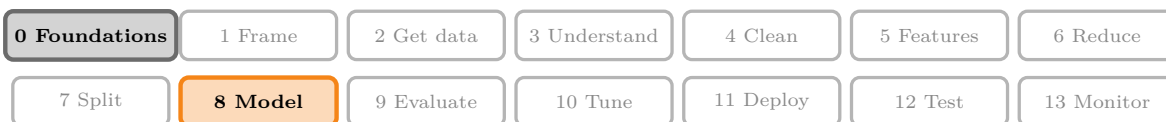


Memoization in Backpropagation

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Where this fits: Pipeline map steps 0 (Foundations), 8 (Model). See the [Course map](#).



- **Builds on:** Derivatives of one variable ([Note 600](#)); Partial derivatives and gradients ([Note 601](#)); Jacobian and matrix gradients ([Note 602](#)); Forward propagation ([Note 1010](#)); Loss functions in deep learning ([Note 1014](#)); Gradient descent ([Note 1017](#)).
- **Leads to:** Improving a neural network ([Note 1021](#)).

1. Overview

Key point: Memoization stores the result of a computation the first time and looks it up afterwards. Backpropagation is the chain rule plus memoization: the derivative at each node is computed once, stored, and reused by every weight behind it.

Memoization is a computer-science technique that speeds up a program by storing the results of expensive function calls and returning the stored result when the same inputs come again. Memoization trades a little memory for a lot of time.

This Note does two things:

1. Shows memoization on its classic example, the Fibonacci numbers.
2. Takes a network with **two** hidden layers, where a first-layer weight reaches the loss along several paths, and shows how storing intermediate derivatives keeps backpropagation fast.

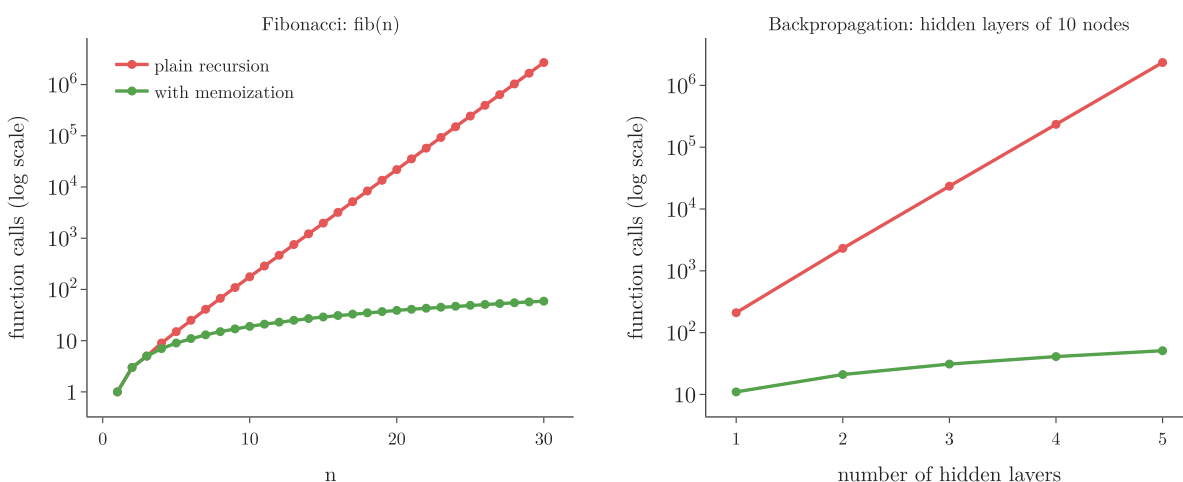


Figure 1: Function calls needed with and without memoization. Left: Fibonacci. Right: backpropagation through hidden layers of 10 nodes

Figure 1 shows the payoff: without memoization the work grows exponentially; with it, almost

linearly.

2. Prerequisites

- The [backpropagation what Note](#) and the [backpropagation how Note](#): the chain-rule derivatives of a one-hidden-layer network.
- The chain rule with several paths: section 6 of the [partial derivatives and gradients Note](#).
- The computation graph: section 10 of the [Jacobian Note](#).

3. Memoization on the Fibonacci numbers

Key point: Plain recursion recomputes the same Fibonacci numbers over and over: 2.7 million calls for $\text{fib}(30)$. Storing each answer in a dictionary cuts that to 59.

3.1 The plain recursive version

Key point: $\text{fib}(n) = \text{fib}(n - 1) + \text{fib}(n - 2)$, with $\text{fib}(0) = \text{fib}(1) = 1$. Short and correct, but very slow.

In the Fibonacci sequence each term is the sum of the two before it: 1, 1, 2, 3, 5, 8, 13, ... Counting from $\text{fib}(0) = \text{fib}(1) = 1$, a recursive function computes it directly.

Python: Fibonacci by recursion.

```
def fib(n):  
    if n == 0 or n == 1:  
        return 1  
    return fib(n - 1) + fib(n - 2)    # the function calls itself
```

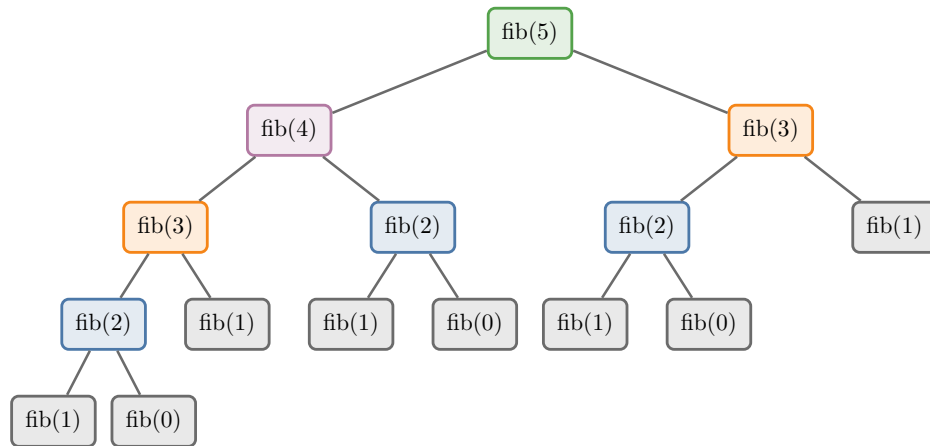
```
fib(5), fib(10), fib(12)           # 8, 89, 233
```

A function that calls itself is **recursive**. Every call for $n \geq 2$ makes two more calls, until they reach 0 or 1.

The recursive version works, but it slows down sharply as n grows: $\text{fib}(30)$ already needs 2,692,537 calls (0.22 seconds), and each further step of n multiplies the work by about 1.6. At $n = 40$ it would take about half a minute; $n = 100$ would take millions of years.

3.2 Why it is slow

Key point: $\text{fib}(5)$ computes $\text{fib}(3)$ twice and $\text{fib}(2)$ three times; for large n the same subtrees are recomputed millions of times.



15 calls in all: fib(3) is computed 2 times, fib(2) 3 times, fib(1) 5 times.
With memoization each value is computed once and then looked up.

Figure 2: Every call made by fib(5): the same values are computed again and again

Figure 2 draws all the calls made by fib(5). fib(5) needs fib(4) and fib(3); fib(4) needs fib(3) again, and so on. fib(3) is computed twice and fib(2) three times: 15 calls for a single answer. For fib(48) the tree would be astronomically large, almost all of it repeats.

Extra: The number of calls is $2 \text{fib}(n) - 1$. Call it $C(n)$: each call makes one call plus the calls of its two children, so $C(n) = 1 + C(n-1) + C(n-2)$ with $C(0) = C(1) = 1$. Then $C(n) + 1$ follows the Fibonacci rule and starts at 2, so $C(n) + 1 = 2 \text{fib}(n)$. The count grows like 1.618^n (the golden ratio). The growth is often loosely called 2^n ; either way it is **exponential time**: each extra step multiplies the work by a constant factor.

3.3 The memoized version

Key point: Keep a dictionary of answers. Before computing fib(n), look it up; after computing it, store it.

Python: Fibonacci with memoization.

```
def fib_memo(n, d):
    if n in d:                    # computed before: look it up
        return d[n]
    d[n] = fib_memo(n - 1, d) + fib_memo(n - 2, d)  # store
    return d[n]
```

```
fib_memo(30, {0: 1, 1: 1})      # 1346269, in 59 calls
```

d is a **dictionary**: it maps each n already computed to its answer. n in d checks whether the key is there.

fib(30) now needs 59 calls instead of 2,692,537, and fib(100) only 199 (Figure 1, left). Each value is computed once; every later request is a lookup. The cost is the memory for the

dictionary, one entry per value. Memoization is the core of **dynamic programming**, a family of algorithms built on reusing solutions to overlapping sub-problems (Cormen et al. 2009, Ch. 15).

Extra: Python's standard library does this in one line: putting `@functools.lru_cache(maxsize=None)` above `def fib(n):` stores every result automatically (Python docs, `functools`).

4. Derivatives in a network with two hidden layers

Key point: In a 3-3-2-1 network, the output-layer weights need one chain, the middle-layer weights a longer one, and the first-layer weights two chains added together, because O_{11} reaches the output through both nodes of layer 2.

4.1 The network

Key point: 3 inputs, hidden layers of 3 and 2 nodes, 1 output: 23 trainable parameters.

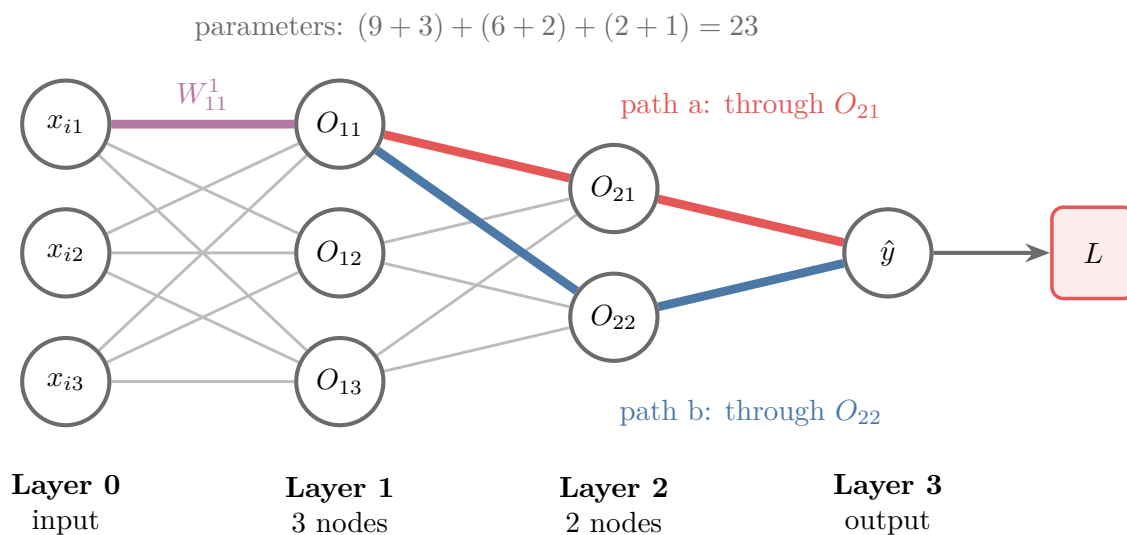


Figure 3: The 3-3-2-1 network. A change in W_{11}^1 changes O_{11} , which reaches $\hat{y}$ along two paths

The [backpropagation what Note](#) used one hidden layer. Figure 3 adds a second: 3 inputs, hidden layers of 3 and 2 nodes, and 1 output. The network has $(9 + 3) + (6 + 2) + (2 + 1) = 23$ trainable parameters (see the [MLP notation Note](#)).

The hidden nodes use the sigmoid, the output is linear, and the loss is $(y - \hat{y})^2$; the same reasoning works for classification. For numbers we take one **observation** (one record, one row of the data table) $x = (0.5, -1, 2)$ with $y = 1$ and fixed random weights. Forward

propagation gives $O_{11} = 0.783$, $O_{21} = 0.374$, $O_{22} = 0.208$ and $\hat{y} = -0.194$, so $\partial L / \partial \hat{y} = -2(1 - (-0.194)) = -2.387$.

4.2 A weight of the output layer

Key point: $\partial L / \partial W_{11}^3 = \partial L / \partial \hat{y} \cdot O_{21}$: one link, as before.

W_{11}^3 connects O_{21} to the output, so

$$\frac{\partial L}{\partial W_{11}^3} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial W_{11}^3} = -2.387 \times 0.374 = -0.893$$

4.3 A weight of the middle layer

Key point: Through O_{21} only: $\partial L / \partial \hat{y} \cdot \partial \hat{y} / \partial O_{21} \cdot \partial O_{21} / \partial W_{11}^2$.

W_{11}^2 connects O_{11} to node O_{21} . Changing it changes O_{21} , which changes $\hat{y}$, which changes L :

$$\frac{\partial L}{\partial W_{11}^2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial O_{21}} \cdot \frac{\partial O_{21}}{\partial W_{11}^2}$$

Here $\partial \hat{y} / \partial O_{21} = W_{11}^3$ and $\partial O_{21} / \partial W_{11}^2 = O_{21}(1 - O_{21}) O_{11}$ (sigmoid slope times input). With the network's numbers this is 0.160. Every one of the 6 middle-layer weights follows the same pattern.

4.4 A weight of the first layer: two paths

Key point: O_{11} feeds both O_{21} and O_{22} , so its effect on the loss is the sum over both paths.

Now take W_{11}^1 . The weight changes O_{11} , but O_{11} goes **forward along two paths** (Figure 3): into O_{21} (path a) and into O_{22} (path b). Both end at $\hat{y}$.

When a variable affects a function through two intermediate variables, the chain rule multiplies along each path and **adds the paths** (see section 6 of the [partial derivatives and gradients Note](#)). For $h(f(x), g(x))$:

$$\frac{dh}{dx} = \frac{\partial h}{\partial f} \frac{df}{dx} + \frac{\partial h}{\partial g} \frac{dg}{dx}$$

Applied here:

1. **In words:** follow each path from L back to W_{11}^1 , multiply the derivatives along it, and add the two products.
2. **Formula:**

$$\frac{\partial L}{\partial W_{11}^1} = \frac{\partial L}{\partial \hat{y}} \left[\frac{\partial \hat{y}}{\partial O_{21}} \frac{\partial O_{21}}{\partial O_{11}} + \frac{\partial \hat{y}}{\partial O_{22}} \frac{\partial O_{22}}{\partial O_{11}} \right] \frac{\partial O_{11}}{\partial W_{11}^1}$$

3. **Example:** path a contributes -0.0110 and path b -0.0028 , so

$$\frac{\partial L}{\partial W_{11}^1} = -0.0110 + (-0.0028) = -0.0138$$

`tf.GradientTape` returns -0.893 , 0.160 and -0.0138 for the three weights, matching the hand formulas (Notebook).

The other 8 first-layer weights need the same two-path sum. With only two hidden layers, one derivative is already a sum of products of five factors. With ten hidden layers, the number of paths from a first-layer node to the output is the product of all the later layer widths: for widths of 10, that is 10^9 paths.

5. Where memoization comes in

Key point: Many of these products share their first factors. Working backwards from the loss, we compute $\partial L / \partial O$ for each node once, store it, and reuse it for every weight behind that node.

5.1 The repeated pieces

Key point: $\partial L / \partial \hat{y}$ appears in all 23 derivatives; $\partial L / \partial O_{21}$ in every weight that reaches the loss through O_{21} .

Look at what the formulas of Section 4 share:

- $\partial L / \partial \hat{y}$ appears in every one of the 23 derivatives.
- $(\partial L / \partial \hat{y})(\partial \hat{y} / \partial O_{21})$, the derivative of the loss with respect to O_{21} , appears in the 3 weights entering O_{21} and in path a of all 9 first-layer weights.
- The bracket of Section 4.4, $\partial L / \partial O_{11}$, appears in all 3 weights entering O_{11} .

Computed naively, each derivative rebuilds these pieces from scratch, exactly as `fib` recomputes `fib(3)`.

5.2 Backpropagation stores one number per node

Key point: $\partial L / \partial O$ for a node = the sum, over the nodes it feeds, of their stored $\partial L / \partial O$ times slope times weight. One pass from the output back gives all of them.

The memoized method keeps one stored number per node: the derivative of the loss with respect to that node's output. Going backwards, layer by layer:

1. **In words:** a node's derivative is built from the stored derivatives of the nodes it feeds; each weight's derivative is then its node's stored derivative times the node's slope times the weight's input.
2. **Formula:** for node j of layer l , with sigmoid slope $s_{l+1,k} = O_{l+1,k}(1 - O_{l+1,k})$ (or 1 for the linear output),

$$\frac{\partial L}{\partial O_{lj}} = \sum_k \frac{\partial L}{\partial O_{l+1,k}} s_{l+1,k} W_{jk}^{l+1}, \quad \frac{\partial L}{\partial W_{ij}^l} = \frac{\partial L}{\partial O_{lj}} s_{lj} O_{l-1,i}$$

3. **Example:** $\partial L / \partial O_{11}$ is computed once, from the stored $\partial L / \partial O_{21}$ and $\partial L / \partial O_{22}$. Then all three weights entering O_{11} reuse it, and the two-path sum of Section 4.4 is never repeated.

Python: Plain recursion versus memoization.

```
from functools import lru_cache
```

```
def dL_dO(l, j):
    # derivative at node j of layer l
    if l == L:
        # the output node
        return -2 * (y - O[L][0])
    return sum(node(l + 1, k) * slope[l + 1][k] * W[l + 1][j, k]
               for k in range(sizes[l + 1]))
```

```
node = dL_dO # plain: recomputes every time
```

```
node = lru_cache(maxsize=None)(dL_dO) # memoized: computes once
```

The recursion calls `node`, so switching `node` between the two lines switches the whole computation between plain and memoized. Both give identical gradients.

5.3 How much work it saves

Key point: For the 3-3-2-1 network: 59 node evaluations without memoization, 6 with. With 5 hidden layers of 10 nodes: 2,345,610 against 51.

Counting how often a node's derivative is evaluated to get all the weight gradients:

Network	Plain recursion	Memoized
3-3-2-1 (Figure 3)	59	6
1 hidden layer of 10	210	11
3 hidden layers of 10	23,410	31
5 hidden layers of 10	2,345,610	51

Each extra hidden layer multiplies the plain count by about 10 (the layer width), while the memoized count grows by 10: one evaluation per node (Figure 1, right). The gap is the same exponential-versus-linear gap as Fibonacci.

The price is memory: the forward pass must keep every node's output, and the backward pass every node's derivative, until the update is done. The small cost in space buys an enormous saving in time.

6. Backpropagation = chain rule + memoization

Key point: The maths is the chain rule; the efficiency is memoization. Keras and TensorFlow implement exactly this on the computation graph.

Backpropagation combines two ideas:

- **The chain rule** (mathematics) says what each derivative is: a sum, over paths, of products of local derivatives.
- **Memoization** (computer science) computes those derivatives efficiently: each shared piece is computed once, stored and reused, working backwards from the loss.

On the computation graph of section 10 of the [Jacobian Note](#), this is the backward pass: keep every intermediate value going forward, then go backward once, multiplying local derivatives and adding where paths meet. Libraries such as Keras and TensorFlow do this automatically, so the whole gradient costs only a small multiple of one forward pass, typically 2 to 3 times (Baydin et al. 2018, §3).

Extra: Storing every activation is why training a network needs much more memory than using it to predict: prediction can throw each layer's outputs away as soon as the next layer is computed, while training must keep them for the backward pass (Chen et al. 2016).

7. Summary

	Fibonacci	Backpropagation
Repeated sub-problem	$\text{fib}(k)$ for smaller k	$\partial L / \partial O$ of each node
Plain recursion	2,692,537 calls for $\text{fib}(30)$	2,345,610 node evaluations, 5 layers of 10
Memoized	59 calls	51 node evaluations
What is stored	one value per n	each node's output (forward) and derivative (backward)

- Memoization stores the results of expensive calls and reuses them: memory for time.
- With two hidden layers, a first-layer weight reaches the loss along several paths, and its derivative is the sum of the path products.
- Many derivatives share the same pieces; computing $\partial L / \partial O$ once per node, from the output backwards, removes all repetition.
- Backpropagation is the chain rule applied with memoization.

8. Sources

- Python documentation, `functools.lru_cache`.
- Cormen, Leiserson, Rivest and Stein, *Introduction to Algorithms*, 3rd ed., MIT Press, 2009, Ch. 15 (dynamic programming; memoization in §15.3).
- Baydin, Pearlmutter, Radul and Siskind, “Automatic Differentiation in Machine Learning: a Survey”, *JMLR*, 2018, §3 (cost and storage of reverse mode).
- Chen, Xu, Zhang and Guestrin, “Training Deep Nets with Sublinear Memory Cost”, arXiv:1604.06174, 2016 (stored activations dominate training memory).

9. Key terms

Term	Meaning
Memoization	Storing the result of a function call and returning the stored result when the same input comes again
Recursion	A function that calls itself on smaller inputs
Exponential time	Work that is multiplied by a constant factor with every step of the input size
Dynamic programming	Solving a problem by reusing the stored solutions of its overlapping sub-problems
Dictionary (Python)	A lookup table from keys to values, written <code>{key: value}</code>
<code>lru_cache</code>	Python's built-in memoization: it stores the results of a function automatically

Term	Meaning
Path (in a network)	A route from a node to the output along connections; a derivative sums its products over all paths